Tata Vishnu Vardhan Rao

Email: tatavishnurao@gmail.com
Mobile: +91-7989073776

Github: github.com/tatavishnurao
LinkedIn: linkedin.com/in/tatavishnurao
X: x.com/tatavishnurao

Experience

- **SochDB** Remote
  Core Contributor – Embedded AI-Native Database *April 2026 - July 2026*
  - **Core Engine Contributions**: Core contributor to SochDB, an embedded **Rust** database unifying **HNSW vector search**, **MVCC**, and **ACID transactions** as a single storage engine for **AI agent memory**, replacing brittle vector-DB + relational-DB stacks with one embeddable primitive.
  - **State-of-the-Art Retrieval Benchmarking**: Owned end-to-end **LoCoMo** retrieval benchmarking, pushing scores to 97.17% **Hit@200** / 94.84% **Recall@200** (84.77%/79.57% @10, 91.50%/86.88% @20) via a candidate-retrieval + **cross-encoder reranking** pipeline, exceeding published results from competing memory-layer startups.
  - **Benchmark Infrastructure & Competitive Positioning**: Built the **benchmark harness** and competitive-analysis tooling behind SochDB's retrieval claims, directly informing founder-level pitch and fundraising materials on retrieval quality vs. competitors.
- **Stealth AI Startup** Remote
  Early Engineering Contributor – AI Agents & Evaluation *May 2025 - August 2025*
  - **LLM Workflow Prototyping**: Built 4 **LLM workflow prototypes** for early-stage AI product experiments, integrating **LLM APIs**, **prompt orchestration**, backend API calls, and application logic for internal MVP validation.
  - **LLM Evaluation**: Developed **DeepEval**-based evaluation scripts covering 120 test cases across 18 prompt and workflow variants, measuring **hallucination**, answer relevance, and task-completion quality during rapid iteration.
  - **Deployment Support**: Containerized and deployed 2 **Next.js** application prototypes using **Docker** and **Azure Container Registry**, reducing repeat demo setup from 7 manual steps to 3 reproducible deployment steps.

Projects

- **tailbench (LLM Inference Benchmarking, Tail Latency, Rust)**: Built an open-loop, **coordinated-omission-safe** latency harness in **Rust** for measuring **tail latency** of **LLM inference** and **GPU serving** under load, with **HdrHistogram** latency timelines, **bootstrap confidence intervals**, and reproducible run manifests. Published `tailbench` (library) and `tailbench-phantom` (a scripted fake OpenAI-compatible server for ground-truth validation) to **crates.io**.
  Tech: Rust, HdrHistogram, Statistics, OpenAI-compatible APIs, Apache-2.0
- **ctxd (Context Runtime for AI Agents, Retrieval, RAG)**: Built a token- and latency-aware context runtime that retrieves candidates (**BM25** + dense), fuses them with deterministic **RRF**, and assembles whole evidence chunks with provenance into one auditable `ContextPacket` within a **token budget**; **Postgres/pgvector** storage, tenant-scoped API, **API-key/JWT auth**, **Prometheus metrics**. Evaluated on **BEIR**: lexical reaches 0.662 **nDCG@10** on SciFact (published **BM25** baseline 0.665); hybrid added +0.069 **recall@100** but not top-rank quality, so lexical is the default and hybrid opt-in.
  Tech: Python, PostgreSQL, pgvector, BM25, Model2Vec, BEIR, Docker
- **SignalFlow-rs (Rust DSP Engine, Audio Feature Extraction, Streaming Systems)**: Built a library-first **Rust** audio DSP crate for end-to-end **log-Mel** feature extraction, with reusable modules for WAV input, framing, windowing, spectrum computation, Mel projection, cached and streaming extraction, stress testing, and **Criterion** benchmarks; binary, examples, and benches share one `signalflow_rs` library API for testability and benchmark consistency.
  Tech: Rust, Cargo, DSP, FFT, Log-Mel Features, Streaming, Criterion

Open Source

- **SGLang PR #46078 (LLM Inference Serving, Rust, gRPC Test Harness)**: Fixed a workspace build break in SGLang's Rust gRPC benchmark harness by adding four missing HiCache RPCs (with `Status::unimplemented` handlers) to the `SglangService` mock, restoring `cargo test --workspace`; validated with workspace/integration tests, Clippy, and `fmt`. Reviewed and merged into `main` by core maintainers.
  Tech: Rust, Cargo, gRPC, Benchmarking, Git, Linux
- **PyTorch AO PR #4332 and AI Infrastructure Contributions**: Contributed prototype bug fixes and code-quality improvements across PyTorch AO modules, including RoPE fusion detection tests, quantization cleanup, and embedding API/export cleanup; also worked on benchmark diagnostics and reproducibility in AI systems evaluation workflows.
  Tech: Python, PyTorch, Quantization, Benchmarking, Git, Linux

Skills Summary

- **Languages** : Rust, Python, C++, SQL
- **ML/AI** : PyTorch, Transformers, DeepEval, LLM Evaluation, Scikit-learn
- **Systems** : Rust DSP, FFT, Log-Mel Features, Streaming Pipelines, Benchmarking
- **Infra** : Docker, Linux, REST APIs, Azure Container Registry, PostgreSQL, MySQL
- **Tools** : Git, GitHub, Cargo, Jupyter
- **Focus** : AI Inference, ML Systems, Agent Memory Systems, Computational Graphics

Education

- **Anurag University** Hyderabad, India
  Bachelor of Technology - Computer Science and Engineering; GPA: 3.26/4.00 *July 2023 - Present*
  Courses: *Operating Systems, Data Structures, Analysis of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases*

Honors and Awards

- Awarded title of Networking Star by Founder of Codebasics - March, 2023

Volunteer Experience

- **Community Lead at Microsoft AI Innovators Hub** Hyderabad, India
  Conducted online and offline technical & soft-skills training impacting over 3000 students. *Jan 2025 - August 2025*

- **Vice President at Malai AU Club** Hyderabad, India
  Organized technical events and delivered workshops reaching over 500 students. *July 2025 - April 2026*